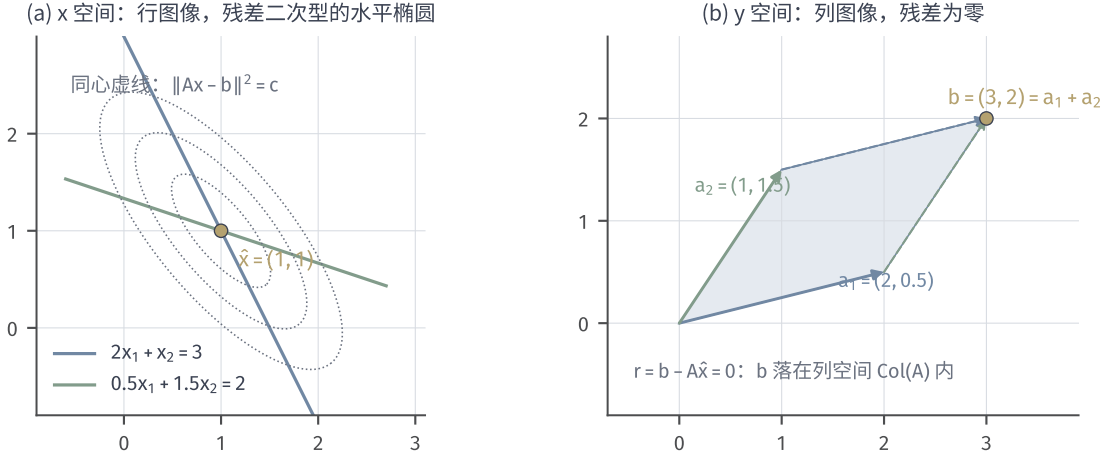
(a) x 空间：行图像，残差二次型的水平椭圆
同心虚线：‖Ax − b‖² = c
x̂ = (1, 1)
2x₁ + x₂ = 3
0.5x₁ + 1.5x₂ = 2
(b) y 空间：列图像，残差为零
b = (3, 2) = a₁ + a₂
a₂ = (1, 1.5)
a₁ = (2, 0.5)
r = b − Ax̂ = 0：b 落在列空间 Col(A) 内